

0xMERIDIAN CONSTELLATION

UC02: Counter-UAS Drone Detection & Interdiction

AI at the Edge — Zero Raw Data Egress — ZK-Proof Verified Detection

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EXECUTIVE SUMMARY

The 0xMeridian Constellation for Counter-UAS (Unmanned Aerial System) operations represents a paradigm shift from centralized, cloud-dependent drone detection systems to a federated, edge-native architecture. This document presents UC02, a comprehensive drone detection and interdiction use case deployed across 5 sovereign nodes, implementing Zero-Knowledge Proof (ZK-Proof) verified detection, multi-sensor fusion, and cryptographic human oversight.

Key Innovation: Raw sensor data (4K video, RF IQ samples, audio waveforms) never leaves the sensor nodes. Only 192-byte mathematical proofs traverse the network, achieving 99.99% bandwidth reduction while maintaining forensic-grade auditability.

1. OPERATIONAL SCENARIO

1.1 Mission Context

Primary Objective: Protection of critical infrastructure (airports, power plants, government facilities) from unauthorized drone intrusions.

Threat Model:

- Hostile UAS reconnaissance or payload delivery
- Swarm attacks with multiple coordinated drones
- RF spoofing and electronic countermeasures
- GPS jamming attempts

Operational Environment:

- Perimeter: 5km radius protected zone
- Response Time: <2 seconds from detection to interdiction
- Compliance: GDPR, EU AI Act, EASA, ISO 27001, STANAG 4586

2. CONSTELLATION ARCHITECTURE

2.1 The Five-Node Federation

The 0xMeridian C-UAS system operates as a federated constellation of 5 autonomous nodes, each running the 0xMeridian AI OS with Magellano inference engine on Apple Silicon.

Node-1: COMMAND HUB (The Brain) **Role:** Central aggregation, global verification, threat assessment, human interface

Capabilities:

- Sensor Fusion Engine: Bayesian correlation of multi-sensor inputs
- ZK Global Verifier: Groth16 proof verification (<2ms per proof)
- Threat Assessment Module: Trajectory prediction, impact zone calculation
- Human Interface: Dashboard with annotated video (not raw), confidence breakdown
- U-Space Reporter: Automated regulatory reporting

Hardware: Apple Silicon M2 Ultra, 64GB RAM, HSM (Hardware Security Module)

Node-2: RADAR-RF SENSOR (The Ear) **Role:** Electromagnetic spectrum monitoring, protocol classification, triangulation

Specifications:

- SDR Wideband Receiver: 2.4-5.8GHz (WiFi/FPV), 700MHz-2.6GHz (LTE/5G)
- Protocol Database: DJI OcuSync (all versions), Crossfire, ExpressLRS, WiFi, Bluetooth
- Direction Finding: Multi-antenna bearing estimation ($\pm 2^\circ$ accuracy)
- Range Estimation: Time-of-flight and RSSI fusion ($\pm 50\text{m}$ at 2km)
- Doppler Analysis: Velocity vector calculation ($\pm 0.5\text{ m/s}$)

Detection Performance:

- FPV drones: 5km range
- WiFi-controlled drones: 10km range
- Classification confidence: >97% for known protocols

Node-3: ELECTRO-OPTICAL (The Eye) **Role:** Visual detection, tracking, classification, size estimation

Specifications:

- PTZ Camera: 4K resolution, 30x optical zoom
- AI Inference: YOLOv8 fine-tuned for UAV detection (mAP@0.5: 0.94)
- Classification: Drone vs bird vs aircraft vs debris
- Tracking: Persistent tracking with Kalman filter (Track ID: TRK-0xXXXX)
- Estimation: Wingspan calculation for drone class identification

Edge Processing:

- Inference latency: <100ms on Apple Silicon Neural Engine
- Frame processing: 30fps continuous
- Bounding box hash: SHA256 for ZK-Proof generation

Node-4: ACOUSTIC ARRAY (The Listener) **Role:** Acoustic signature detection, direction of arrival, motor fingerprinting

Specifications:

- Microphone Array: 8x MEMS microphones, beamforming configuration
- Spectral Analysis: FFT-based harmonic detection
- Database: 200+ drone motor signatures (brushless, brushed, turbine)
- DoA Accuracy: $\pm 5^\circ$ via beamforming algorithms
- Range: 500m (quiet motors) to 1km (loud motors)

Signature Library:

- DJI Mavic series (all models)
- DJI Phantom series
- FPV racing drones
- Custom-built multirotors
- Fixed-wing UAVs

Node-5: C-UAS EFFECTOR (The Shield) **Role:** Active countermeasures, safe landing protocol, kinetic fallback

Countermeasure Suite:

1. RF Jamming (Selective):

- Targeted jamming on 2.4GHz and 5.8GHz control channels
- Protocol-aware: jams only drone link, preserves other communications
- Power: 10W EIRP per band

2. GPS Spoofing:

- Coordinated GPS signal injection
- Redirects drone to predefined safe landing zone
- Coordinates: Configurable (default: 41.9028, 12.4964)

3. Safe Landing Protocol:

- Monitors drone failsafe behavior
- Ensures controlled descent (not crash)
- Verifies touchdown confirmation

4. Kinetic Net Launcher (Fallback):

- Deployable for RF-resistant drones
- Range: 100m
- Net capture with parachute recovery

Safety Features:

- Human override mandatory (ADR-026)
- Geofencing: No jamming in civilian air corridors
- Power limits: ITU Radio Regulations compliant

3. ZK-PROOF MESSAGING ARCHITECTURE

3.1 Core Principle: Zero Raw Data Egress

Traditional C-UAS systems stream raw sensor data to the cloud:

- 4K video: ~20 Mbps per camera
- RF IQ samples: ~50 Mbps
- Audio: ~1 Mbps
- **Total: ~100 Mbps continuous**

0xMeridian approach:

- Edge inference generates ZK-Proofs locally
- Only 192-byte proofs transmitted
- **Total: ~10 Kbps (99.99% reduction)**

3.2 ZK-Proof Structure: RF_DETECTION_EVENT (192 bytes)

```
ZK-PROOF: RF Drone Detection Event
-----
Origine: RADAR-RF (Node-2)
Destination: COMMAND HUB (Node-1)
Protocol: Bus-4 (gRPC mTLS) + WireGuard Mesh

Payload (192 bytes):
├─ detection_type: RF_EMISSION
├─ frequency_band: 2.4GHz (WiFi control channel)
├─ signal_strength: -62 dBm
├─ protocol_class: DJI_OcuSync_v3 (confidence: 0.97)
├─ bearing: 045° ± 2°
├─ estimated_range: 2.3 km
├─ doppler_velocity: +15 m/s (approaching)
├─ timestamp_proof: Generation within T-500ms
├─ sensor_attestation: HSM Ed25519 signature
└─ zk_proof: Groth16 proof (correct classification)

Verification (Node-1):
1. Verify Groth16 proof < 2ms
2. Cross-check bearing with optical track (Node-3)
3. Validate HSM attestation chain
4. If valid + confidence > 0.95 → ALERT: DRONE DETECTED
5. If invalid → Circuit Breaker (ADR-007) → log anomaly
```

3.3 ZK-Proof Structure: OPTICAL_TRACK (192 bytes)

```
ZK-PROOF: Optical Drone Track
-----
```

Origine: ELECTRO-OPTICAL (Node-3)
Destination: COMMAND HUB (Node-1)

Payload:

- └─ detection_type: VISUAL_TRACK
- └─ object_class: UAV_MULTIROTOR (confidence: 0.94)
- └─ bounding_box_hash: sha256(bbox_coordinates)
- └─ track_id: TRK-0x7A3F (persistent across frames)
- └─ position_estimate: AZ 044° EL 12° (camera-relative)
- └─ velocity_vector: [+14.2, +0.5, -1.1] m/s
- └─ size_estimate: wingspan ~0.35m (micro-drone class)
- └─ frame_merkle_root: root of last 30 frames (1s window)
- └─ inference_proof: Groth16 (YOLOv8 execution correctness)

Verification:

1. Verify Groth16 proof < 2ms
2. Correlate with RF bearing (Node-2): $|\Delta\theta| < 5^\circ$
3. Correlate with acoustic direction (Node-4)
4. Fused confidence > 0.95 → TRACK CONFIRMED
5. Update threat assessment → route to C-UAS Effector

3.4 ZK-Proof Structure: ACOUSTIC_HIT (192 bytes)

ZK-PROOF: Acoustic Drone Detection

Origine: ACOUSTIC ARRAY (Node-4)
Destination: COMMAND HUB (Node-1)

Payload:

- └─ detection_type: ACOUSTIC_SIGNATURE
- └─ rotor_profile: QUAD_BRUSHLESS (frequency: 8.2kHz ± 200Hz)
- └─ harmonic_fingerprint_hash: sha256(spectral_features)
- └─ direction_of_arrival: AZ 046° ± 5°
- └─ estimated_distance: 450m ± 50m
- └─ snr: 22 dB (above ambient noise floor)
- └─ classification: DJI_Mavic_3_class (confidence: 0.88)
- └─ acoustic_proof: Groth16 (spectral analysis correctness)

Multi-Sensor Correlation (Node-1):

1. RF bearing: 045° | Optical: 044° | Acoustic: 046°
2. Angular spread $\Delta\theta = 2^\circ$ → HIGHLY CORRELATED
3. Fused classification: DJI_Mavic_3 (conf: 0.993)
4. THREAT LEVEL: 0.87 → ALERT → Request Human Confirmation

3.5 ZK-COMMAND: INTERDICTION_ORDER

ZK-COMMAND: Interdiction Order (ADR-026 Override)

Origine: COMMAND HUB (Node-1) + Human Operator
Destination: C-UAS EFFECTOR (Node-5)

```
Command Structure:
├─ header:
│   ├── command_type: INTERDICTION_ORDER
│   ├── threat_id: THR-0x7A3F
│   ├── priority: 240 (critical)
│   └─ ttl: 60 seconds
├─ target:
│   ├── classification: DJI_Mavic_3
│   ├── position: AZ 045° EL 12° Range 450m
│   ├── velocity: [+14.2, +0.5, -1.1] m/s
│   └─ predicted_impact_zone: ZONE_ALPHA (15s ETA)
├─ action:
│   ├── primary: RF_JAMMING_SELECTIVE (2.4GHz + 5.8GHz)
│   ├── secondary: GPS_SPOOFING (redirect to safe zone)
│   ├── fallback: KINETIC_NET (if RF-resistant)
│   └─ safe_zone: GPS coord [41.9028, 12.4964] (landing)
└─ authentication:
    ├── operator_id: OPR_ROSSI_2201
    ├── hsm_signature: Ed25519 (Secure Enclave)
    ├── biometric_hash: sha256(fingerprint_template)
    └─ chain_of_command: [CMDR → OPR → SYSTEM] verified
```

4. COMPLETE MESSAGE CATALOG (36 Types)

Detection Messages (5 types)

Message	Size	Flow	Description
RF_DETECTION_EVENT	192B	Node-2 → Node-1	RF emission detected, protocol classified
RF_TRIANGULATION	256B	Node-2 → Node-1	Multi-antenna bearing + range estimate
OPTICAL_DETECTION	192B	Node-3 → Node-1	Object detected in video frame (YOLOv8)
OPTICAL_TRACK_UPDATE	192B	Node-3 → Node-1	Continuous tracking with velocity vector
OPTICAL_CLASSIFICATION	192B	Node-3 → Node-1	Drone type classification (model ID)

Acoustic Messages (3 types)

Message	Size	Flow	Description
ACOUSTIC_HIT	192B	Node-4 → Node-1	Rotor acoustic signature detected
ACOUSTIC_DIRECTION	128B	Node-4 → Node-1	Direction of arrival from mic array
ACOUSTIC_FINGERPRINT	192B	Node-4 → Node-1	Motor fingerprint matched to drone DB

Fusion Messages (3 types)

Message	Size	Flow	Description
FUSION_CORRELATION	384B	Node-1 internal	Multi-sensor data correlated for track
FUSION_CONFIDENCE	128B	Node-1 → All	Fused confidence score for active threat
THREAT_ASSESSMENT	256B	Node-1 → All	Threat level + predicted trajectory

Interdiction Messages (5 types)

Message	Size	Flow	Description
INTERDICTION_REQUEST	384B	Node-1 → Node-5	Request to engage with countermeasures
INTERDICTION_CONFIRM	192B	Node-5 → Node-1	Countermeasure activated confirmation
INTERDICTION_RESULT	256B	Node-5 → Node-1	Outcome: neutralized/lost/escaped
JAMMING_ACTIVE	128B	Node-5 → All	RF jamming on specific frequency band
GPS_SPOOF_ACTIVE	192B	Node-5 → All	GPS spoofing directing to safe zone

Human Override Messages (2 types)

Message	Size	Flow	Description
HUMAN_OVERRIDE_CMD	384B	Operator → Node-1	Human override command (ADR-026)
HUMAN_OVERRIDE_ACK	192B	Node-1 → All	Override acknowledged and propagated

Gossip Protocol Messages (4 types)

Message	Size	Flow	Description
GOSSIP_THREAT_MAP	800B	Any → Any (k=3)	Current threat map + coverage state
GOSSIP_NODE_HEALTH	256B	Any → Any (k=3)	CPU/GPU/temp/uptime status
GOSSIP_COVERAGE_MAP	512B	Any → Any (k=3)	Sensor coverage zones per node
MERKLE_AUDIT_BATCH	224B	Node-1 → Archive	Batch of 100 events Merkle root

System Messages (10 types)

Message	Size	Flow	Description
CIRCUIT_BREAKER_TRIP	128B	Any → Node-1	Anomalous behavior detected, node isolated
CIRCUIT_BREAKER_RESET	128B	Node-1 → Any	Node re-admitted after verification
SENSOR_CALIBRATION	512B	Node-1 → Sensors	Calibration parameters update
DECOY_ALERT	256B	Node-1 → All	Possible spoofing/decoy detected
SAFE_MODE_ENTER	128B	Any → All	Node entering safe mode (degraded)

Table 7 – continued

Message	Size	Flow	Description
SAFE_MODE_EXIT	128B	Any → All	Node exiting safe mode (nominal)
AMBASSADOR_HANDSHAKE	1024B	Any Any	Initial mesh join + mutual TLS auth
AMBASSADOR_KEEPALIVE	64B	Any Any	Periodic liveness check
PARTITION_DETECTED	128B	Any → Local	Network partition detected
PARTITION_RESOLVED	256B	Any → All	Partition healed, reconciliation start

External Messages (2 types)

Message	Size	Flow	Description
U_SPACE_REPORT	512B	Node-1 → External	Report to UTM/U-Space authority
LAW_ENFORCEMENT_ALERT	384B	Node-1 → External	Alert to law enforcement API

5. OPERATIONAL SEQUENCE: STEP-BY-STEP

Phase 1: Multi-Sensor Detection (T+0 — T+500ms)

Step 1: RF Detection (T+0ms)

- Node-2 SDR scanner detects anomalous RF emission at 2.4GHz
- Pattern matching identifies DJI OcuSync v3 protocol (confidence: 0.97)
- Bearing estimated: $045^{\circ} \pm 2^{\circ}$
- Range: 2.3km
- Doppler: +15 m/s (approaching)
- Magellano generates ZK-Proof (192 bytes) of classification
- Proof sent to Node-1 via Bus-4

Step 2: Optical Detection (T+100ms)

- Node-3 PTZ camera receives cue from gossip mesh (bearing 045°)
- Slew-to-cue automatic activation
- YOLOv8 detects aerial object: class UAV_MULTIROTOR, confidence 0.94
- Track ID assigned: TRK-0x7A3F
- Bounding box hash and velocity vector calculated
- ZK-Proof (192 bytes) sent to Node-1

Step 3: Acoustic Detection (T+200ms)

- Node-4 microphone array detects acoustic signature at 8.2kHz fundamental (typical brushless quad motor)

- Beamforming calculates DoA: $046^{\circ} \pm 5^{\circ}$
- Fingerprint match: DJI Mavic 3 class (confidence 0.88)
- ZK-Proof (192 bytes) sent to Node-1

Step 4: Initial Verification (T+400ms)

- Node-1 receives 3 independent ZK-Proofs
- Verifies Groth16 of each (<2ms each)
- Initiates multi-sensor correlation process

Phase 2: Sensor Fusion & Threat Assessment (T+500ms — T+1s)

Step 5: Angular Correlation

- Node-1 executes angular correlation:
- RF: 045°
- Optical: 044°
- Acoustic: 046°
- Angular spread: $2^{\circ} \rightarrow$ HIGHLY CORRELATED
- Single target confirmed

Step 6: Classification Fusion

- Bayesian fusion of classifications:
- RF $\rightarrow$ DJI OcuSync (0.97)
- Optical $\rightarrow$ Multirotor (0.94)
- Acoustic $\rightarrow$ Mavic 3 (0.88)
- Fused result: DJI Mavic 3, confidence 0.993

Step 7: Trajectory Prediction

- Velocity vector: $[+14.2, +0.5, -1.1]$ m/s
- ETA to protected perimeter: 15 seconds
- Estimated impact: ZONE_ALPHA (critical area)

Step 8: Threat Assessment

- THREAT_LEVEL = 0.87
- Automatic threshold (0.98) not reached
- Human confirmation required
- Alert sent to operator: *"DRONE DETECTED: DJI Mavic 3, 450m, approaching ZONE_ALPHA, ETA 15s. AUTHORIZE INTERDICTION?"*

Phase 3: Human Decision & Interdiction (T+1s — T+3s)

Step 9: Human Authorization

- Operator receives alert on dashboard with:
- Position map
- Annotated video feed (bounding box overlay, not raw)
- Confidence breakdown by sensor
- Response options
- Operator confirms: AUTHORIZE RF_JAMMING + GPS_SPOOF

Step 10: Command Generation

- Node-1 generates INTERDICTION_ORDER signed with HSM (Ed25519)
- Includes: operator_id, biometric_hash, chain_of_command verification
- Command sent to Node-5 via Bus-4 with priority 240

Step 11: Countermeasure Activation

- Node-5 receives order
- Verifies HSM signature and chain of command
- Activates selective RF jammer on 2.4GHz + 5.8GHz
- Simultaneously activates GPS spoofing to redirect drone to safe landing zone

Step 12: Confirmation

- Node-5 confirms: INTERDICTION_ACTIVE
- Jamming on target frequencies active
- GPS spoof active
- Continuous monitoring: drone loses link with remote operator, enters failsafe mode, begins descent to safe zone

Phase 4: Monitoring & Outcome (T+3s — T+30s)

Step 13: Descent Tracking (Optical)

- Node-3 tracks drone during descent
- Velocity vector changes: deceleration, controlled descent to safe zone
- Track status: LANDING

Step 14: Descent Confirmation (Acoustic)

- Node-4 confirms RPM motor reduction (pitch decrease)
- Consistent with landing profile

Step 15: Landing Confirmation (T+25s)

- Drone lands in safe zone

- Node-3 confirms: object on ground, no movement
- Node-4 confirms: motors off (no acoustic signature)
- Node-5 deactivates jamming and GPS spoof

Step 16: Threat Resolution

- Node-1 generates THREAT_RESOLVED
- Threat level: 0.00
- Interdiction outcome: NEUTRALIZED_SAFE_LANDING
- Audit entry with Merkle root

Phase 5: Post-Incident & Forensics (T+30s+)

Step 17: Incident Report

- Node-1 compiles automatic incident report including:
 - Complete timeline
 - All ZK-Proofs (verifiable)
 - Human decision (signed)
 - Outcome
- Report sent to: U-Space/UTM authority, law enforcement, insurance

Step 18: Mesh Propagation

- Gossip mesh propagates THREAT_CLEARED to all nodes
- Coverage map returns to nominal
- All nodes confirm operational status

Step 19: Forensic Archive

- Merkle Tree updated with final batch
- Root hash committed to persistent store
- Complete audit trail available for:
 - Regulatory inspection
 - Legal proceedings
 - Insurance claims
 - Operational review

6. GOSSIP SWIM PROTOCOL

Protocol Specification (ADR-004)

Protocol: Gossip SWIM (Scalable Weakly-consistent Infection-style Process Group Membership)

Parameters:

- T_gossip: 200ms — each node contacts k=3 random peers every 200ms
- T_suspect: 600ms (3 periods) — after 3 missed responses, node suspected
- Indirect Probe: m=3 intermediate nodes attempt contact before declaring FAILED
- Convergence: $O(\log N)$ hops — for N=5, convergence in ~7 hops = <1.4s

Message Structure:

```
GOSSIP_THREAT_MAP_UPDATE
header:
  msg_type: THREAT_MAP_UPDATE
  origin_node: 0xM-Node-{1-5}
  sequence: [monotonic counter]
  timestamp: [unix epoch microseconds]
payload:
  active_threats: [{id, class, pos, vel, threat_level}...]
  sensor_coverage_map: {zone: [sensor_types]}
  node_health: {cpu, mem, gpu, temp, uptime}
  interdiction_status: {active_jamming: bool, target_id}
verification:
  zk_digest: sha256(zk_proof_of_state_computation)
  signature: Ed25519(node_key)
  merkle_root: root_of_last_100_events
```

Payload Size: ~800 bytes per gossip message

7. DISASTER RECOVERY SCENARIOS

Scenario A: Sensor Node Compromise (Physical Destruction)

Trigger: Node-2 (RF Sensor) physically destroyed or compromised

Timeline:

- T+0ms: Gossip SWIM detects missed response (T_suspect=600ms, 3 periods)
- T+600ms: Indirect probe: 3 nodes attempt direct contact with Node-2
- T+1800ms: Node-2 declared FAILED. Mesh reconfigures
- T+2000ms: Coverage gap compensation:
 - Node-3 (Optical) increases FOV and scan rate
 - Node-4 (Acoustic) extends detection radius
- T+5000ms: Alert to Command Hub: “RF SENSOR OFFLINE — Degraded Mode Active”
- Threat detection continues with 2 sensors (confidence threshold increased to 0.98)

Recovery:

- Spare node deployment or mobile RF sensor unit
- Automatic reconnection via Ambassador Agent + mutual TLS

- State synchronization via Merkle diff

Scenario B: Network Partition (Split-Brain)

Trigger: WireGuard mesh partitions: {Node-1, Node-3} vs {Node-2, Node-4, Node-5}

Partition Alpha (1,3): Command Hub + Optical

- Can detect and classify visually
- Cannot interdict (Effector in Partition Beta)
- Enters Safe Mode: logging only, no action

Partition Beta (2,4,5): RF + Acoustic + Effector

- Can detect via RF/Acoustic
- Can interdict
- Without Command Hub supervision
- Enters Safe Mode: interdiction disabled, requires local human confirmation

Reconciliation:

- Upon connectivity restoration, Merkle Tree diff identifies divergent events
- Conflict resolution via timestamp ordering + ZK-Proof verification
- No data loss: all events cryptographically preserved

Scenario C: Adversarial Spoofing Attack

Trigger: Attacker emits false RF signals to simulate non-existent drone

Detection:

- Multi-sensor fusion fails:
- RF detects “drone” at bearing 090°
- Optical and Acoustic: no correlation
- Single-sensor detection with confidence <0.60 classified as POSSIBLE_DECOY

Response:

- Alert: “SPOOFING ATTEMPT DETECTED — Single-sensor anomaly at bearing 090°”
- Node-2 activates frequency hopping analysis to identify spoofing transmitter
- Transmitter location estimated via triangulation
- Coordinates sent to law enforcement

Countermeasure:

- No interdiction triggered (insufficient confidence)
- Operator notified of potential electronic attack
- System continues monitoring with heightened alert

8. REAL-TIME PERFORMANCE METRICS

Metric	Value	Notes
RF Detection Range	5 km (FPV) / 10 km (WiFi)	< 500ms latency
Radar Detection	3 km (micro-drone RCS 0.01m²)	< 200ms latency
Optical Classification	2 km (4K camera + AI)	< 100ms (edge inference)
Acoustic Detection	500m (quiet motors)	< 300ms latency
Sensor Fusion Confidence	> 99.2%	3-sensor correlation
Fusion Latency	< 50ms	Bayesian computation
ZK-Proof Generation	192 bytes per event	< 100ms (GPU) / 500ms (CPU)
ZK-Proof Verification	O(1) –always constant	< 2ms
Jamming Response Time	Target lock to neutralization	< 2s (automated) / 5s (human-in-loop)
False Positive Rate	< 0.1%	After multi-sensor fusion
Gossip Convergence	O(log N) = 7 hops for 5 nodes	< 1.4s full mesh

All metrics are mathematically verified via ZK-Proofs. They are not trust-based estimates but cryptographically verifiable proofs.

9. REGULATORY COMPLIANCE

Regulation	Compliance Mechanism
GDPR Art. 5-6	ZK-Proof guarantees no raw video data leaves sensor nodes. Only mathematical proofs travel. Privacy by Design.
EU AI Act Art. 6, 14	High-risk AI system with mandatory Human Override (ADR-026). Classification subject to audit.
EASA Reg. 2019/947	U-Space integration for remote drone identification. Coordination with aviation authorities.
ISO 27001	Cryptographic immutable audit trail (Merkle + ZK). Every detection event has verifiable signature.
STANAG 4586	NATO C-UAS standard compatibility. Interoperable interface with allied defense systems.
ITU Radio Reg.	RF jamming operates only on authorized drone frequencies. Spectral geofencing to avoid civilian interference.

10. SECURITY & PRIVACY GUARANTEES

Data Sovereignty (ADR-030)

Raw sensor flows (4K video, RF raw IQ samples, audio waveforms) NEVER leave the peripheral nodes (2,3,4). They remain in local Shared Memory (Bus 3) and are processed by Magellano inference. Only ZK-Proofs (192 bytes) travel on the network.

Zero-Trust Verification (ADR-013)

The Command Hub (Node-1) does not trust sensor nodes. It mathematically verifies every claim (e.g., “I detected a DJI Mavic at 2.3km bearing 045”) via ZK-Proof before activating any countermeasure. A compromised sensor cannot deceive the system.

Human Override (ADR-026)

Automatic interdiction only above threat level 0.98. Below this threshold, ALWAYS human confirmation with cryptographic signature. Verifiable chain of command: Operator → Commander → System.

Anti-Spoofing (Multi-Sensor)

Spoofing attack on single sensor detected by lack of correlation with other 2 sensors. Confidence remains below threshold → no interdiction on false positive.

Resilience (ADR-007)

If a node fails, mesh reconfigures in 5 seconds. Coverage gap compensated by other sensors with increased threshold. Safe Mode guarantees operational continuity.

Forensic Integrity

Immutable Merkle Tree with ZK-Proofs. Every detection, decision, and action is cryptographically verifiable by third parties (authorities, courts, insurers) without accessing sensitive sensor data.

11. COMPARATIVE ANALYSIS: 0xMeridian vs Cloud-Centric

Parameter	0xMeridian C-UAS	Cloud-Centric C-UAS
Detection Latency	< 500ms edge (multi-sensor)	1-5s cloud roundtrip + processing
Interdiction Latency	< 2s (automated) / 5s (human)	10-30s (cloud decision + command)
Privacy (Sensor Data)	Zero data egress. 192 bytes proof	Streaming video/RF to cloud (Gbps)
False Positive Rate	< 0.1% (3-sensor fusion)	2-5% (single sensor cloud)
Required Bandwidth	~10 Kbps (proof + gossip)	~100 Mbps (video + RF streams)
Resilience	Mesh auto-healing in 5s	Single point of failure (cloud)
Audit Trail	Merkle + ZK (math-verified)	Log files (trust-based)
GDPR Compliance	Native (no PII egress)	Requires DPA, DPIA, consent
Infrastructure Cost	5 edge nodes (~50K€ total)	Cloud + networking (~200K€/year)
Scalability	N nodes, O(log N) convergence	Linear with cloud cost

12. WHY THIS CHANGES THE GAME FOR ENTERPRISE

Automatic GDPR Compliance

No DPA required for video processing because video is never transmitted or centrally stored. Only the mathematical proof of the event travels. Privacy by Design.

Deterministic Latency

Processing is edge-based (<500ms detection, <2s interdiction). A cloud system would have at least 1-5s of roundtrip + processing, unacceptable for critical threats.

99.99% Bandwidth Reduction

Instead of streaming 4x 4K cameras + RF IQ + audio raw to cloud (hundreds of Mbps), stream only ~10 Kbps of proofs + gossip.

Incontestable Audit

Every detection and interdiction has a cryptographically verifiable signature by third parties (aviation authorities, insurers, courts) without accessing sensitive data. Fundamental for liability.

Operational Resilience

Self-healing mesh system. No single point of failure. If cloud goes down, cloud-centric system is blind. If one 0xMeridian node goes down, others compensate.

13. CONCLUSION

The 0xMeridian Constellation for Counter-UAS operations demonstrates how distributed, edge-native AI systems can transform critical infrastructure protection. By combining:

- **Zero-Knowledge Proofs** for verifiable privacy
- **Multi-sensor fusion** for <0.1% false positive rates
- **Cryptographic human oversight** for regulatory compliance
- **Federated architecture** for resilience

UC02 provides a blueprint for next-generation security systems that do not compromise between privacy, performance, and accountability.

“The Constellation does not ask for trust. It asks for verification.”
— ADR-030, 0xMeridian Federated Constellation Governance

APPENDIX A: GLOSSARY

Term	Definition
ZK-Proof	Zero-Knowledge Proof: cryptographic method to prove a statement is true without revealing the underlying data
Groth16	Efficient ZK-Proof system with constant-size proofs (~200 bytes) and fast verification
HSM	Hardware Security Module: tamper-resistant hardware for key storage and cryptographic operations
Merkle Tree	Cryptographic data structure enabling efficient verification of large datasets
Gossip SWIM	Protocol for distributed failure detection and membership in unreliable networks

Table 12 – continued

Term	Definition
DoA	Direction of Arrival: technique to determine the direction of a signal source
U-Space	European framework for safe and secure drone operations in low-altitude airspace
ADR	Architecture Decision Record: documented decision with context and consequences

APPENDIX B: REFERENCE DOCUMENTS

- ADR-001: Rust Kernel Implementation
- ADR-003: Magellano HAL Specification
- ADR-004: Gossip SWIM Protocol
- ADR-007: Resilience & Safe Mode
- ADR-012: WireGuard Mesh Networking
- ADR-013: Zero-Trust Verification
- ADR-026: Human Override Requirements
- ADR-030: Data Sovereignty & Constellation Governance

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